

Liquid chromatography under critical conditions: Practical applications in the analysis of amphiphilic polymers

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Received: 7 October 2008 / Revised: 15 December 2008 / Accepted: 18 December 2008 / Published online: 24 January 2009
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Abstract Liquid chromatography under critical conditions (LCCC) allows the separation of block copolymers from the corresponding homopolymers as well as the separation of homopolymers according to their functionality. At the transition of exclusion and adsorption mode, the polymer chain becomes “chromatographically invisible,” and thus a separation according to other structural units can be achieved. In the case of block copolymers this situation can be utilized to detect and determine unwanted homopolymers. At critical conditions for the repeat unit, the other block may be eluted in the exclusion or the adsorption regime. In the first case, the block copolymer is eluted before the homopolymer, and its molar mass can be determined as in size-exclusion chromatography. In the second case, it is eluted later than the homopolymer, and the separation of the individual oligomers can be achieved. Advantages and limitations of different approaches in LCCC are discussed.

Keywords Liquid chromatography · Critical conditions · Block copolymers · Functional polymers

Introduction

In liquid chromatography the separation mechanism for polymers depends on the interaction parameter c , which describes the interaction of the structural unit with the stationary phase [1]. This parameter is negative in size-exclusion chromatography (SEC), in which retention decreases with an increase in molar mass, and is positive in liquid adsorption chromatography (LAC), where retention increases exponentially with an increase in the number of repeat units.

In both cases, the elution volume V_e depends on the coefficient K for the distribution of the polymer between the interstitial volume V_i and the pore volume V_p :

$$V_e = V_i + KV_p. \quad (1)$$

This equation is well known in SEC (where K can assume values between 0 and 1): a polymer of very high molar mass (above the exclusion limit) is eluted at V_i (as $K=0$), while small molecules (which have access to the entire pore volume) are eluted at the void volume:

$$V_0 = V_i + V_p \text{ (as } K = 1 \text{)}.$$

In LAC, the distribution coefficient can assume very high values (much larger than 1).

The distribution coefficient is related to the change in free energy related to the transition of the polymer from the interstitial volume into the pores of the stationary phase:

$$\ln K = \Delta G = \Delta H - T\Delta S. \quad (2)$$

In general, the driving force in SEC is the entropy change, while in LAC retention is governed by the enthalpic interaction of the repeat unit with the stationary phase.

Presented at the 9th International Symposium on Instrumental Analysis, Pécs, Hungary, July 29 to August 2, 2008.

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Table 1 Possible situations in liquid chromatography under critical conditions (LCCC) of amphiphilic block copolymers

Column	Mobile-phase polarity	Hydrophilic (EO)	Hydrophobic (PO, BO, HO, CL)
RP	High	LCCC	LAC
RP	Low	SEC	LCCC
NP	High	LCCC	SEC
NP	Low	LAC	LCCC

EO ethylene oxide, PO propylene oxide, BO butene oxide, HO hexene oxide, CL caprolactone, RP reversed phase, NP normal phase, SEC size-exclusion chromatography, LAC liquid adsorption chromatography

At the transition point of these two mechanisms, which is often termed the “critical adsorption point” (CAP), there is a special mobile-phase composition (at a given temperature), where the interaction parameter has a value of zero: under such conditions, the polymer is eluted—independent of its molar mass—at the void volume of the column. In other words, at the CAP the polymer chain becomes chromatographically invisible [2–5].

This situation can be utilized in the characterization of functional oligomers, block and graft copolymers, as it allows the separation of polymer homologues according to other structural units, such as their end groups or other blocks [3, 6–11] (M.I. Malik, B. Trathnigg, and C.O. Kappe, unpublished results).

Especially interesting are amphiphilic polymers, which contain a hydrophilic block [such as poly(ethylene oxide)] linked to a hydrophobic block [such as poly(propylene oxide), a higher poly(alkylene oxide), and polycaprolactone, PCL]. These materials have applications in many fields, such as nonionic surfactants and emulsifiers, in which the hydrophilic (water-soluble) polyoxyethylene chain is linked to lipophilic (oil-soluble) groups or blocks. Hence, these products are surface-active: they arrange themselves at the surface of water

or at the oil–water interface and thus lower the surface tension. Amphiphilic block copolymers of ϵ -caprolactone and ethylene oxide (EO) have special properties [12]. As the poly(ethylene glycol) (PEG) block minimizes platelet deposition onto surfaces exposed to the bloodstream [13], they can be used to produce blood-contacting implant devices having low protein adsorption, membranes for separation processes, biosensors, chromatographic supports, contact lenses, and in blood and protein storage applications, etc. [14]. These polymers may contain various impurities (such as the homopolymers, residual monomer, and initiator), which may influence the properties of the products. Consequently, there is a strong need for reliable characterization methods. Owing to the complex nature of amphiphilic polymers, this will also require complex techniques, such as two-dimensional liquid chromatography. There have also been attempts to apply pyrolysis gas chromatography [15, 16] and mass spectroscopy [mainly matrix-assisted laser desorption/ionization time of flight (MALDI-TOF) mass spectroscopy] [17–19]. Pyrolysis gas chromatography is very useful in the analysis of cross-linked polymers, but it cannot be considered as a standard technique in the characterization of block copolymers: since 2000, there have been very few papers in

Table 2 Critical conditions for different repeat units on different columns in different mobile phases

Reversed phase				Repeat unit					
Stationary phase		Mobile phase	Composition (wt%)	EO	PO	BO	HO	CL	n-Alkyl
C18	Various brands	Methanol-water	80–90	CAP	LAC	LAC	Ins.	LAC	
		Acetone-water	26–27	CAP	LAC	Ins.			
			~ 90	CAP	CAP	LAC			
		THF-water	80–85	SEC	CAP	LAC		CAP	LAC
			87–92	SEC		CAP	LAC	SEC	LAC
	Phenyl PEG Poly-DVB		90–95	SEC			CAP	SEC	LAC
		Spherisorb S5P	~ 90	LAC	CAP	SEC	Ins.		
		Discovery HS PEG	44	SEC	CAP	LAC	Ins.		LAC
		Jordi	92	SEC	CAP	LAC			
		THF-water	80	SEC	CAP	LAC		CAP	LAC
Normal phase	Diol		85	SEC		CAP	LAC	SEC	LAC
			87	SEC			CAP	SEC	LAC
		Nucleosil 100-5 OH	78	CAP	SEC				

PEG poly(ethylene glycol), DVB divinylbenzene, THF tetrahydrofuran, CAP critical adsorption point, Ins. insoluble

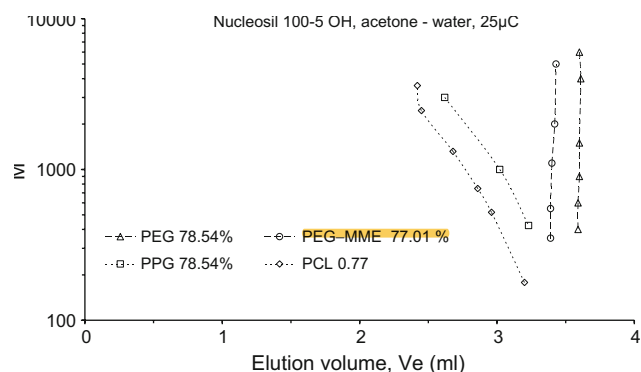


Fig. 1 Elution of poly(ethylene glycol) (PEG), PEG–monomethyl ether (PEG–MME), poly(propylene glycol) (PPG), and polycaprolactone (PCL) on Nucleosil 100-5 OH in acetone–water (77 and 78.5 wt%) at the critical adsorption point (CAP) for ethylene oxide (EO)

the literature. MALDI-TOF mass spectroscopy is problematic in the analysis of highly polydisperse samples; it works very well, however, with samples with a narrow molar mass distribution. When applied to fractions from a chromatographic separation, it yields highly valuable information.

In liquid chromatography, one may use different separation modes, which separate species according to different criteria (corresponding to the interaction parameters of the components), and combine them to achieve a two-dimensional separation.

In the case of block copolymers, the interaction parameters c_A and c_B of the individual structural units (A and B) will typically have different values.

If both c_A and c_B are negative, the overall molar mass distribution can be determined by SEC. In SEC, retention decreases with increasing molar mass and shows only a minor influence of chemical composition. It must be

mentioned that the SEC calibration functions of different polymers can be considerably different (as may be the interaction parameters). In this case the chemical composition along the peak is required, which can be obtained from dual detection [20–22].

If both c_A and c_B are positive, a reasonable separation by LAC (with retention increasing exponentially with the number of repeat units) can only be achieved if one block is monodisperse [6, 23]. If, however, both blocks are polydisperse and $c_A \neq c_B$, several superimposed series of peaks will be obtained, and these cannot be resolved.

If one of the interaction parameters for one repeat unit is positive and the other one is negative, each polymer homologue is eluted in SEC order, but far behind the void volume of the column. This mechanism, which is called “liquid exclusion–adsorption chromatography” (LEAC) [24, 25], can be utilized in the separation of monofunctional oligomers with a monodisperse adsorbing block or end group. In the analysis of block copolymers, LEAC may only be applied as the second dimension in two-dimensional chromatography.

If one of the interaction parameters (c_A or c_B) is zero, the corresponding block becomes “chromatographically invisible” and separation according to the other block can be achieved (liquid chromatography under critical conditions, LCCC).

Depending on the interaction parameter of the other repeat unit, this separation may follow a SEC [7, 26–28] (situation A) or a LAC [6, 9, 29, 30] mechanism (situation B).

There could be different options, as has been discussed previously [31, 32] (see Table 1).

On reversed-phase columns, there may be critical conditions for the hydrophilic block in a mobile phase with rather high polarity. In such a mobile phase the hydrophobic block will be eluted in LAC mode. On the other hand,

Fig. 2 Liquid chromatography under critical conditions (LCCC) (at the CAP for EO) of two commercial PCL diols and PEG-MME 1100 on Nucleosil 100-5 OH in 77 wt% acetone. *MW* molecular weight

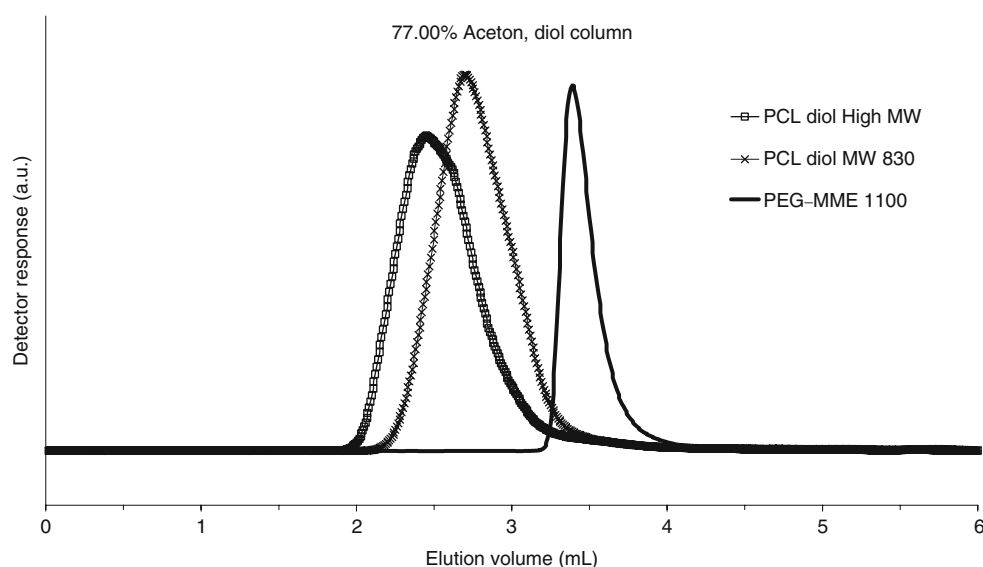
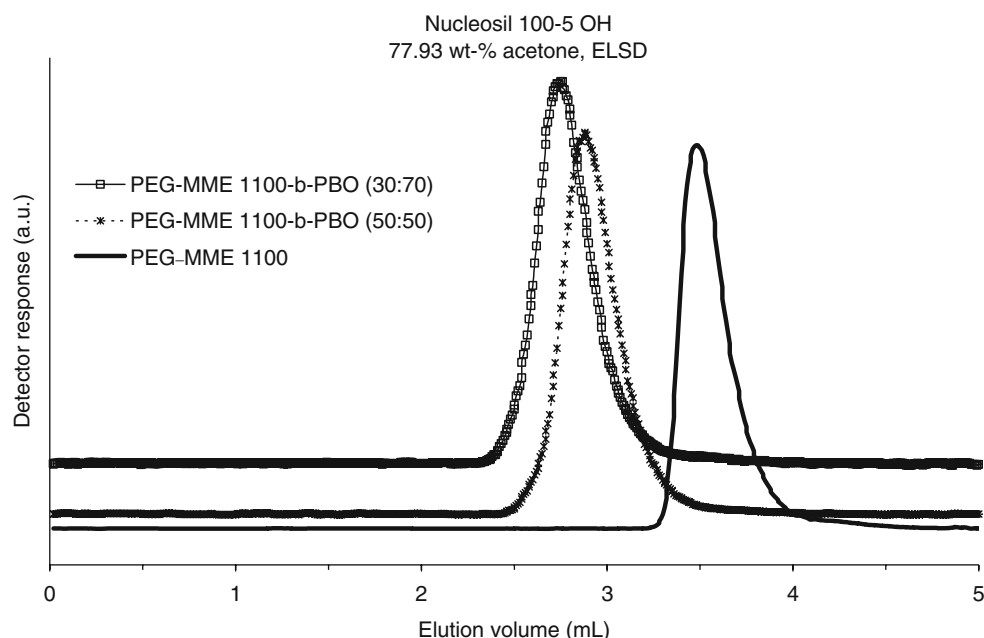


Fig. 3 LCCC (at the CAP for EO) of PEG-MME and the corresponding diblock copolymers with butene oxide (BO) (exclusion conditions for the BO block). *PBO* poly(butene oxide)



critical conditions may also be found for the hydrophobic block in a rather nonpolar mobile phase. In this case, the hydrophilic block will be eluted in SEC mode.

On normal-phase columns, a CAP for the hydrophilic block may be found in a polar mobile phase, in which the hydrophobic block will be eluted in SEC mode. In a mobile phase with low polarity, a CAP may exist for the hydrophobic block: in this case the hydrophilic block will be eluted in LAC mode.

In this study these possible situations were evaluated with respect of advantages and limitations. As practical examples we chose various block copolymers based on epoxide—EO, propylene oxide (PO), butene oxide (BO), hexene oxide (HO), lactone (caprolactone, CL)—repeat units or polymers with terminal alkyl groups. These samples were analyzed on various columns of different nature and in different mobile phases.

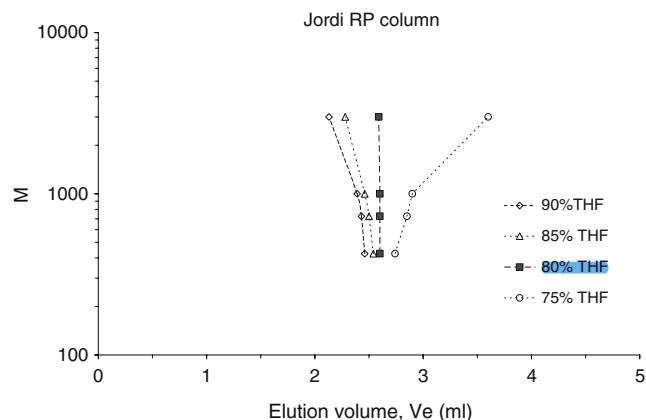


Fig. 4 Critical conditions for PPG on the Jordi RP column in tetrahydrofuran (THF)–water

Materials and methods

PEGs, poly(propylene glycol)s (PPGs), and PCL diols were purchased from Sigma-Aldrich, while homopolymers and block copolymers of higher epoxides as well as block copolymers of PO were synthesized by microwave-assisted anionic ring opening polymerization, as has been described in detail elsewhere [6, 7, 11].

These investigations were performed using a DDS70 (CHROMTECH, Graz, Austria) density detector (according to the mechanical oscillator principle) in all chromatographic systems. Data acquisition and processing was performed using the software program CHROMA, which has been developed for the DDS70 density detector.

The column(s) and density cells were placed in a thermostatted box, in which a temperature of 25.0 °C was maintained for all measurements. The mobile phase was delivered by an ISCO 2350 high performance liquid

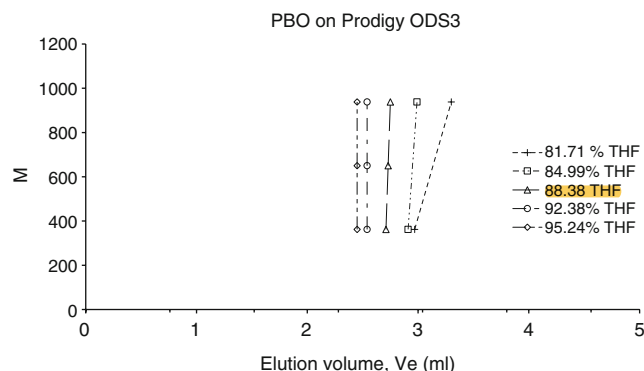


Fig. 5 Critical conditions for PBO on the Prodigy ODS3 column in THF–water

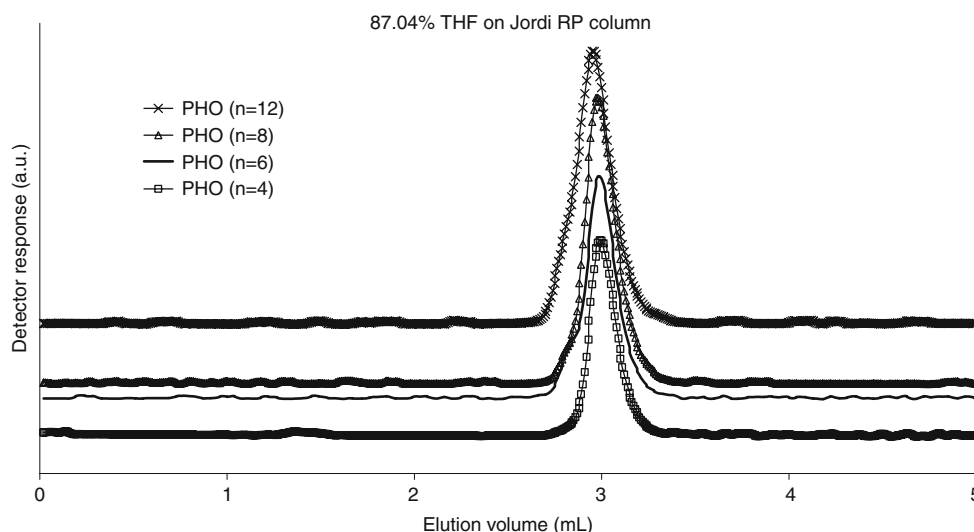


Fig. 6 Overlay of chromatograms of poly(hexene oxide) (PHO) having different average number of repeat units, as obtained on the Jordi column at the CAP for hexene oxide (HO) (87 wt% THF–water)

chromatography (HPLC) pump (ISCO, Lincoln, NE, USA) at a flow rate of 0.5 ml/min. Samples were injected using a Spark SPH 125 Fix autosampler (Spark Holland, Emmen, The Netherlands) equipped with a 20- μ l loop. An RI 2414 (Waters) or a SEDEX 45 ELSD (Sedere, France) refractive index detector was connected to the DDS70 density detector. Nitrogen was used as a carrier gas, and the pressure at the nebulizer was set to 1.0 bar. The evaporator temperature was 30 °C.

The measurements in this study were performed on the following columns:

- **Spherisorb S5X C18** (PhaseSeparations): silica-based octadecyl phase; 250 mm $\times$ 4.6 mm; particle diameter 5 μ m; nominal pore size 300 Å

- **Spherisorb S5P** (PhaseSeparations): silica-based phenyl phase; 250 mm $\times$ 4.6 mm; particle diameter 5 μ m; nominal pore size 80 Å
- **Novapak C18** (Waters, Milford, MA, USA): silica-based octadecyl phase; 300 mm $\times$ 3.9 mm; particle diameter 4 μ m; nominal pore size 60 Å
- **Prodigy ODS3** (Phenomenex, Torrance, CA, USA): silica-based octadecyl phase; 250 mm $\times$ 4.6 mm; particle diameter 5 μ m; nominal pore size 100 Å
- **Chromolith Performance RP-18e 100-4.6** (Merck, Darmstadt, Germany): silica-based monolith with octadecyl groups; 100 mm $\times$ 4.6 mm
- **Jordi Gel DVB 500 RP** column (Jordi, Bellingham, MA, USA): 100% polydivinylbenzene; 250 mm $\times$ 4.6 mm; particle diameter 5 μ m; pore size 500 Å

Fig. 7 LCCC on the Jordi column at the CAP for HO (87 wt% THF–water) of PHO and two diblock copolymers with PEG-MME (size exclusion chromatography conditions for EO)

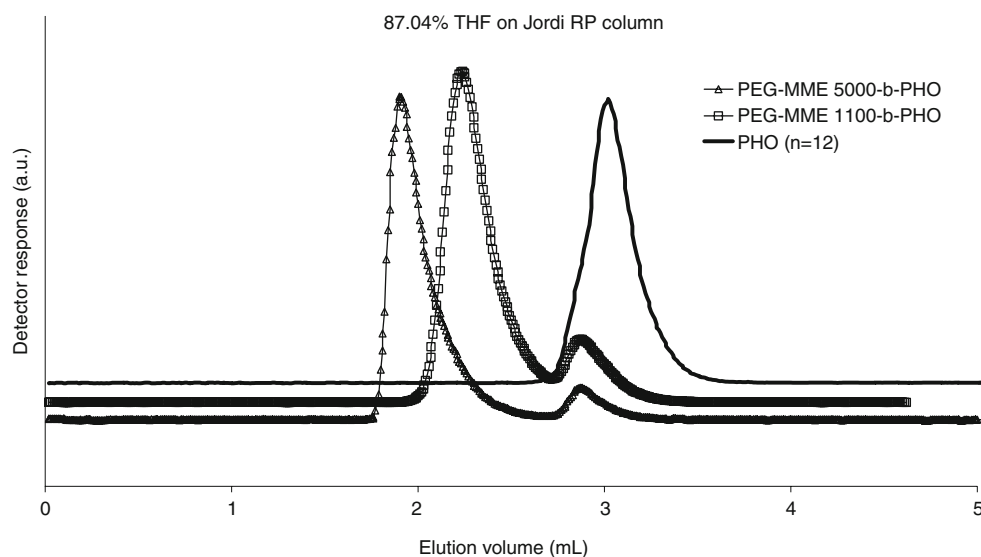
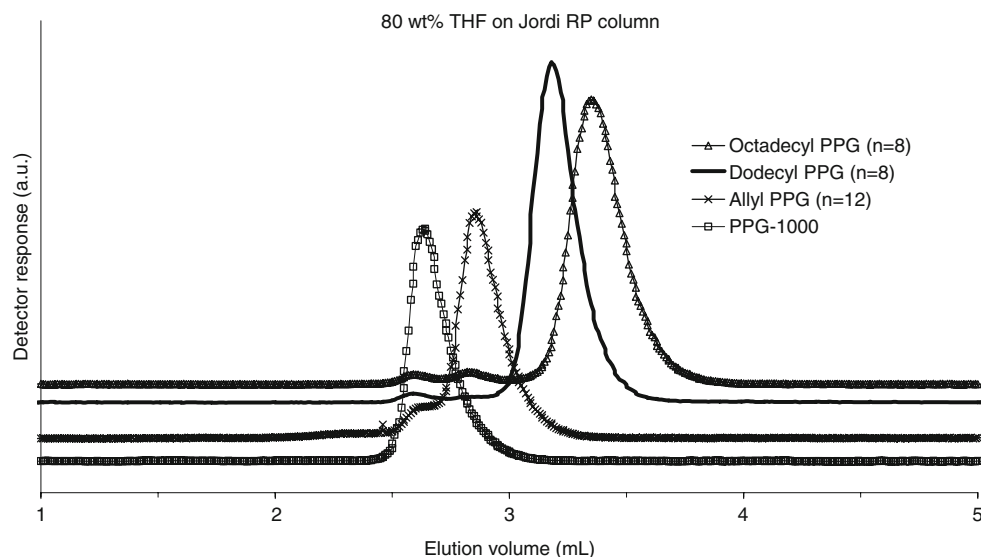


Fig. 8 LCCC (at the CAP for propylene oxide, PO) of PPG diols, monoallyl PPG, and two PPGs with adsorbing end groups



- **Nucleosil 100-5 OH 5 μm** (Macherey-Nagel, Dueren, Germany): silica-based diol phase; 250 mm $\times$ 4.6 mm; particle diameter 5 μm ; pore diameter 100 $\text{\AA}$
- **Discovery HS PEG** (Supelco, Bellefonte, PA, USA): silica-based PEG phase; 250 mm $\times$ 4.6 mm; particle diameter 5 μm ; nominal pore size 120 $\text{\AA}$

The solvents [methanol, acetone, tetrahydrofuran (THF) and water, all HPLC grade] were purchased from Roth (Karlsruhe, Germany). Mobile phases were mixed by mass and vacuum-degassed; their composition was controlled by density measurement using a DMA 60 density meter equipped with a DMA 602 M measuring cell (A. Paar, Graz, Austria).

Results and discussion

The applicability of the situations described in “Introduction” depends on the nature of the samples to be characterized. In Table 2, critical conditions for different repeat units and the elution behavior for other repeat units at the same mobile-phase composition are summarized for all the columns used in this study. Evidently, situation A (the other repeat unit is excluded from pores and is eluted in SEC order) is the more versatile approach, as it can be applied in a much wider range of molar mass (provided that the noncritical block is large enough to be separated from the critical one). On the other hand, situation B (the other repeat unit has strong interaction with the stationary phase, and hence the block is eluted in LAC order) may allow the

Fig. 9 LCCC at the CAP for PO and liquid adsorption chromatography (LAC) conditions for EO (Spherisorb S5P, 90 wt% methanol–water)

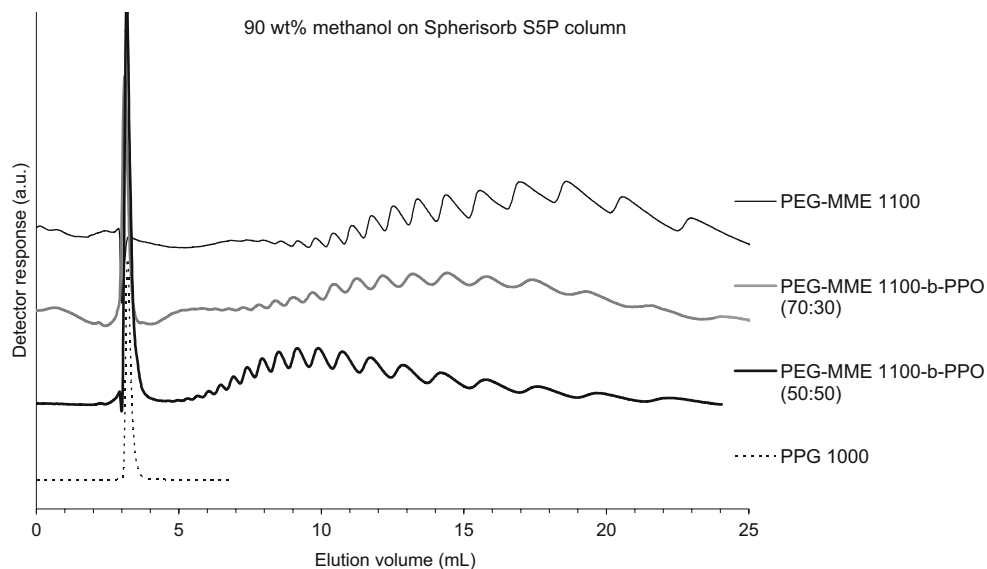
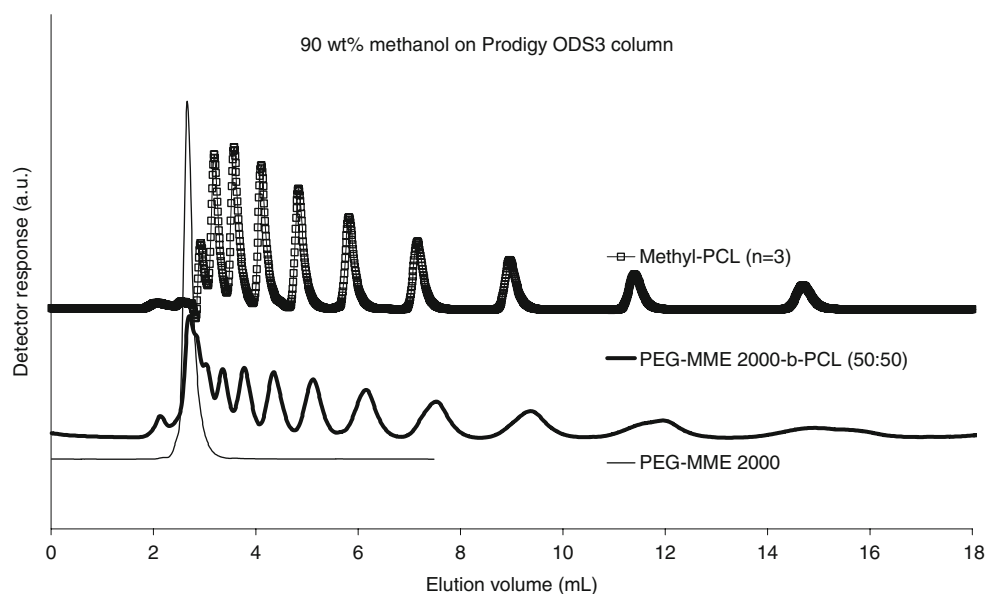


Fig. 10 LCCC at the CAP for EO and LAC conditions for caprolactone (CL) (Prodigy ODS3, 90 wt% methanol–water): PEG-MME 2000, monomethoxy-PCL, and an EO-CL diblock copolymer



separation of the individual oligomers of the noncritical block, but this will work only in a narrow range of molar mass.

Let us first consider situation A. At the CAP for PEG on a normal-phase column (Nucleosil 100-5 OH in 78 wt% acetone), PPG is eluted in SEC mode, and so are other hydrophobic blocks [such as poly(butene oxide) (PBO), poly(hexene oxide) (PHO), and PCL]. This is depicted in Fig. 1.

Figure 2 shows a comparison of the chromatograms of two commercial PCL diols with different molar masses and PEG–monomethyl ether (MME) 1100 which were obtained on the Nucleosil 100-5 OH column in 78 wt% acetone (CAP for EO, SEC for PCL). The molar mass is specified by the producer only for one of the PCL diols ($M_n=830$). We analyzed these samples by SEC on a PLgel Mixed E column (300 mm×7.8 mm, from Polymer Laboratories) in CHCl_3 and found the following values: “PCL diol high MW,” $M_w=3,188$, $M_n=1,706$; “PCL diol MW 830,” $M_w=1,550$, $M_n=843$. Evidently, both PCL samples are clearly separated from PEG-MME, and their elution follows a SEC mechanism.

Under the same conditions, EO–BO diblock copolymers can be separated from PEG-MME 1100, which was used as an initiator in their synthesis (Fig. 3). The elution volume of the block copolymers reflects the length of the BO block.

Critical conditions for PO (at SEC conditions for EO) can be realized on the Jordi column in 90 wt% acetone, as has already been described previously [31, 32], but also in THF–water.

For PBO, a CAP exists on the Jordi column in 84.5 wt% THF (Fig. 4), and on the Prodigy column in above 90 wt% THF (Fig. 5). It is interesting to note that at very high THF concentrations PBO no longer goes in the exclusion regime

but is more or less critical (from 88 to 95 wt% THF); this may be due to a change of the column behavior in eluents with a high organic solvent content, where the influence of residual silanol groups increases dramatically. In such a mobile phase, silica-based reversed-phase columns start behaving as normal-phase columns [33].

For PHO, a CAP is observed on the Jordi column at 87 wt% THF, as can be seen in Fig. 6. In such a mobile phase various PHOs with different degrees of polymerization are eluted at the same volume.

Under these conditions, EO–HO diblock copolymers can be separated from the homopolymer of HO. As can be seen in Fig. 7, the copolymers are eluted in SEC order, and their elution volume reflects the length of the EO block (the initiator is PEG-MME).

At the CAP for the repeat unit of the chain, one may also separate polymers according to their functionality. Figure 8 shows a comparison of the chromatograms (obtained on the Jordi column in 80 wt% THF–water) of a PPG diol, a monoallyl PPG (which may be formed by chain transfer to the monomer), and two monoalkyl PPGs with different alkyl end groups. Evidently, the PPGs initiated with a fatty alcohol are clearly separated from the PPG diol, and it is even possible to discriminate PPG diols and allyl PPGs.

A completely different situation is observed on the Spherisorb S5P column in 90 wt % methanol–water: in such a mobile phase PPG is eluted at the CAP, while PEG is eluted in LAC mode (see Fig. 9)! Under such conditions a separation according to the number of EO units (regardless of the length of the PO block) can be achieved. It must be mentioned that the Spherisorb S5P column (which can normally be considered as a reversed-phase column) behaves in such a mobile phase like a normal-phase column. This seems to be due to the residual silanol

groups on this stationary phase. On other phenyl phases (with a more complete end capping) such a separation could not be achieved.

The combination of critical conditions and LAC can also be utilized in the analysis of EO–CL block copolymers: **on the Prodigy column a CAP for EO exists at 90 wt% methanol.**

In such a mobile phase, PCL oligomers can be resolved to the baseline. As can be seen in Fig. 10, the chromatograms of a PCL initiated with methanol and a diblock copolymer (initiated with PEG-MME 2000) show exactly the same peak pattern. The focus of our current work in this context is to fractionate all these amphiphilic block copolymers using LCCC (at the CAP for one block) and to analyze these fractions by using SEC, LAC, and MALDI-TOF in the second dimension. The results thus obtained will be communicated soon.

Conclusion

In this study we have shown that LCCC can be used for analysis of block copolymers to detect unwanted homopolymer if it is formed during synthesis. The situation in which one block is critical while the other is excluded from the pores can be applied in the analysis of block copolymers with a wide range of molar mass. On the other hand, the situation with an adsorbing end group and critical conditions for the chain is more favorable in the characterization of functional polymers and oligomers. This approach can be applied to block copolymers only in a narrow range of molar mass.

Acknowledgements M.I.M. and H.A. thank the Higher Education Commission (HEC) of Pakistan for a Ph.D. Scholarship. We also thank C. Oliver Kappe of CDLMC for his cooperation in the synthesis of most of the products used for this study.

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